EECE 306 Lab 2 notebook. Coordinate Systems, Points and Vector Components

Team 3: Amelia and Carissa

Fill in every TODO, run `publish('Lab02_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

Contents

Setup

```
clear; close all;
rng(306);
```

Point roundtrips, 1000 random points

```
r = randn(1000,3);
esph = max(em.vec.mag(em.coord.sph2c(em.coord.c2sph(r)) - r));
ecyl = max(em.vec.mag(em.coord.cyl2c(em.coord.c2cyl(r)) - r));
fprintf('spherical   roundtrip max error = %.3e\n', esph);
fprintf('cylindrical roundtrip max error = %.3e\n', ecyl);
```

```
spherical   roundtrip max error = 6.357e-15
cylindrical roundtrip max error = 2.979e-15
```

Poles and the phi convention

```
disp('c2sph([0 0 1])  ='); disp(em.coord.c2sph([0 0 1]))
disp('c2sph([0 0 -1]) ='); disp(em.coord.c2sph([0 0 -1]))
q = [ 1 1 0; -1 1 0; -1 -1 0; 1 -1 0 ];
p = em.coord.c2cyl(q);
fprintf('phi in the four quadrants = %.4f %.4f %.4f %.4f rad\n', p(:,2));
fprintf('all inside [0, 2*pi) = %d\n', all(p(:,2) >= 0 & p(:,2) < 2*pi));
```

```
c2sph([0 0 1])  =
     1     0     0

c2sph([0 0 -1]) =
```

```
    1.0000    3.1416         0

phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
```

Vector component roundtrips and the invariant

```
A  = randn(1000,3);
As = em.coord.vecC2Sph(A, r);
ev = max(em.vec.mag(em.coord.vecSph2C(As, r) - A));
em_ = max(abs(em.vec.mag(As) - em.vec.mag(A)));
fprintf('vector roundtrip (spherical pair) max error = %.3e\n', ev);
fprintf('magnitude change under conversion max       = %.3e\n', em_);
Ac = em.coord.vecC2Cyl(A, r);
ev2 = max(em.vec.mag(em.coord.vecCyl2C(Ac, r) - A));
fprintf('vector roundtrip (cylindrical pair) max error = %.3e\n', ev2);
```

```
vector roundtrip (spherical pair) max error = 1.088e-15
magnitude change under conversion max       = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16
```

The two position demonstration

The same Cartesian vector A = 2x + 3y, expressed in spherical components at two different positions.

```
A0 = [2 3 0];
disp('at (0, 5, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [0 5 0]))
disp('at (4, 0, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [4 0 0]))
% TODO two to four sentences. The vector did not change. Explain exactly
% what did change, and why a position argument is therefore mandatory for
% component conversion and meaningless for point conversion.
%
% The directions of the spherical components (r_hat, theta_hat, phi_hat)
% changed at each position along the vector, since they depend on the polar
% angle theta and the azimuthal angle phi. Spherical axes are local to a
% point and rotate as the point moves through space. Thus, component
% conversion requires a position argument to determine the orientation of
% the local axes in 3D space. Point conversion does not change the
% orientation of the axes; it only changes how the location of the point is
% described. Therefore, a position argument is unnecessary in this case.
```

```
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000    0.0000   -2.0000

at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000    0.0000    3.0000
```

Test Output

```
PASS   test_lab01            0.05 s
PASS   test_lab02            0.03 s
```

```
------------------------------------------------------------
2 passed, 0 failed, 2 total
All tests passing.
```

Interpretation

TODO what class of bug does the roundtrip test catch, and what class does it miss. Name the invariant that catches what the roundtrip misses, and say why it is reference free.

1. The roundtrip test catches the class of bug in which a function treats a cylindrical coordinate (rho, phi, z) as a Cartesian coordinate (x, y, z). The roundtrip test misses the class of bug in which atan2 returns a negative phi when x is negative. 2. The invariant that catches what the roundtrip misses is the magnitude of the vector. It is reference-free because it is unchanged by any component conversion and can always be calculated if the scalar components are known.

Problems encountered

TODO honest account, or NONE. Amelia: Minor syntax and labeling errors when copying similar code between toolkit files. em.coord functions return Nx1 vector outputs, not Nx3. Roundtrip errors from conversions were considerably high. For test 6, phi only shows as one value, not four as intended. Most of these issues were fixed by correcting the way the function takes the individual coordinates from the input. Carissa: Not expressing point and vector conversions as empty Nx3 arrays and as separate or A values, which stopped the random 1000 x 3 test the eight functions test from working properly. No single output for the point conversion function which lead to another error which interfered with matrix multiplication.

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
------------------------------------------------------------
24.2.0.3212159 (R2024b) Update 9
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0   = 2.9979245808e+08 m/s
eta0 = 376.730314 ohm
Derived constant checks passed.
mag([3 4 0])  = 5   (expect 5 exactly)
unit([3 4 0]) =
    0.6000    0.8000         0

unit([0 0 0]) (expect all NaN) =
   NaN   NaN   NaN

angle perpendicular = 1.5707963267949  (expect pi/2)
angle antiparallel  = 3.14159265358979, real = 1  (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
     1     2     3

mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
  PASS  test_lab01           0.02 s
spherical   roundtrip max error = 6.357e-15
cylindrical roundtrip max error = 2.979e-15
c2sph([0 0 1])  =
     1     0     0
```

```
c2sph([0 0 -1]) =
    1.0000    3.1416         0

phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
vector roundtrip (spherical pair) max error = 1.088e-15
magnitude change under conversion max       = 8.882e-16
vector roundtrip (cylindrical pair) max error = 6.280e-16
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000    0.0000   -2.0000

at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000    0.0000    3.0000

  PASS  test_lab02            0.02 s
   r (m)    relative error
    1.0000         0
    2.0000         0
    3.0000         0
    4.0000         0
    5.0000         0
    6.0000         0
    7.0000         0
    8.0000         0
    9.0000    0.0000
   10.0000         0

max |Ex| on bisector      = 0.000e+00 V/m
max |E| on the same plane = 6.631e+02 V/m
|E(r)| / |E(2r)| = 8.0003   (expect near 8 for a dipole)
em.field.E on 5000 points took 0.004 s  (requirement, under 2 s)
  PASS  test_lab03            0.35 s
--------------------------------------------------------------
  3 passed, 0 failed, 3 total
  All tests passing.
```